$(1,-1,3)$ in the direction of $\bar{i}+2\bar{j}+3\bar{k}$

Sol:- Given surface is $2xy+z^2$ at $(1,-1,3)$

let $\phi = 2xy+z^2$

1(A) $\nabla\phi = \bar{i}\frac{\partial\phi}{\partial x} + \bar{j}\frac{\partial\phi}{\partial y} + \bar{k}\frac{\partial\phi}{\partial z}$

$= \bar{i}\frac{\partial(2xy+z^2)}{\partial x} + \bar{j}\frac{\partial}{\partial y}(2xy+z^2) + \bar{k}\frac{\partial}{\partial z}(2xy+z^2)$

$= \bar{i}(2y) + \bar{j}(2x) + \bar{k}(+2z)$

$= 2y\bar{i} + 2x\bar{j} + 2z\bar{k}$

$\nabla\phi_{(1,-1,3)} = -2\bar{i} + 2\bar{j} + 6\bar{k}$

And also given the direction of $\bar{i}+2\bar{j}+3\bar{k}$

let $\bar{a} = \bar{i}+2\bar{j}+3\bar{k}$

$|\bar{a}| = \sqrt{1+4+9} = \sqrt{14}$

Directional derivative $= \nabla\phi \cdot \hat{a} = \nabla\phi \cdot \frac{\bar{a}}{|\bar{a}|}$

$= (-2\bar{i}+2\bar{j}+6\bar{k})\frac{(\bar{i}+2\bar{j}+3\bar{k})}{\sqrt{14}}$

$= \frac{-2+4+18}{\sqrt{14}} = \frac{20}{\sqrt{14}}$

Find div $\bar{F}$ where $\bar{F}$ = gradient $(x^3+y^3+z^3-3xyz)$.

Sol:- Given that $x^3+y^3+z^3-3xyz$

this is also written as $\nabla(x^3+y^3+z^3-3xyz)$.

1(B). $= \left(\bar{i}\frac{\partial}{\partial x}+\bar{j}\frac{\partial}{\partial y}+\bar{k}\frac{\partial}{\partial z}\right).(x^3+y^3+z^3-3xyz)$

$= \bar{i}\frac{\partial}{\partial x}(x^3+y^3+z^3-3xyz)+\bar{j}\frac{\partial}{\partial y}(x^3+y^3+z^3-3xyz)+$

$\bar{k}\frac{\partial}{\partial z}(x^3+y^3+z^3-3xyz)$

$\bar{F} = \bar{i}(3x^2-3yz)+\bar{j}(3y^2-3xz)+\bar{k}(3z^2-3xy)$

div $\bar{F} = \nabla\cdot\bar{F}$

$= \left(\bar{i}\frac{\partial}{\partial x}+\bar{j}\frac{\partial}{\partial y}+\bar{k}\frac{\partial}{\partial z}\right)\left(\bar{i}(3x^2-3yz)+\bar{j}(3y^2-3xz)+\right.$

$\left.\bar{k}(3z^2-3xy)\right)$

$= \frac{\partial}{\partial x}(3x^2-3yz)+\frac{\partial}{\partial y}(3y^2-3xz)+\frac{\partial}{\partial z}(3z^2-3xy)$

$= 6x+6y+6z = 6(x+y+z)$.

2(a)

* Find the angle between $x^2+y^2+z^2=9$, $z=x^2+y^2-3$ at $(2,-1,2)$.

given surfaces $x^2+y^2+z^2=9$.

Let $\phi_1 = x^2+y^2+z^2-9$.

$\phi_2 = x^2+y^2-z-3$.

$$\phi_1 = \bar{i}\frac{\partial\phi_1}{\partial x} + \bar{j}\frac{\partial\phi_2}{\partial y} + \bar{k}\frac{\partial\phi_1}{\partial z}$$

$$= \bar{i}\frac{\partial}{\partial x}(x^2+y^2+z^2-9) + \bar{j}\frac{\partial}{\partial y}(x^2+y^2+z^2-9) + \bar{k}\frac{\partial}{\partial z}$$

$(x^2+y^2+z^2-9)$.

$\nabla\phi_1 = \bar{i}(2x) + \bar{j}(2y) + \bar{k}(2z)$.

$\nabla\phi_1$ at $(2,-1,2) = 4\bar{i} - 2\bar{j} + 4\bar{k}$.

$\nabla\phi_2 = \bar{i}\frac{\partial\phi_2}{\partial x} + \bar{j}\frac{\partial\phi_2}{y} + \bar{k}\frac{\partial\phi_2}{\partial z}$.

$= \bar{i}\frac{\partial}{\partial x}(x^2+y^2-z-3) + \bar{j}\frac{\partial}{\partial y}(x^2+y^2-z-3) + \bar{k}\frac{\partial}{\partial z}(x^2+y^2-z-3)$.

$\nabla\phi_2 = \bar{i}(2x) + \bar{j}(2y) + \bar{k}(-1)$.

$\nabla\phi_2$ at $(2,-1,2) = 4\bar{i} - 2\bar{j} - \bar{k}$.

$|\nabla\phi_1| = \sqrt{16+4+16} = \sqrt{36} = 6$.

$|\nabla\phi_2| = \sqrt{16+4+1} = \sqrt{21}$.

angle b/w two surfaces.

$\cos\theta = \frac{\nabla\phi_1 \cdot \nabla\phi_2}{|\nabla\phi_1|,|\nabla\phi_2|}$

$= \frac{(4\bar{i} - 2\bar{j} + 4\bar{k})(4\bar{i} - 2\bar{j} - \bar{k})}{6\sqrt{21}}$.

$= \frac{16+4-4}{6\sqrt{21}}$.

$\cos\theta = \frac{16}{6\sqrt{21}} = \frac{8}{3\sqrt{21}}$

$\theta = \cos^{-1}\left(\frac{8}{3\sqrt{21}}\right)$.

which is required solution.

If $\bar{F} = xy^2\bar{i} + 2x^2yz\bar{j} - 3yz^2\bar{k}$ then find diverge $\bar{F}$ at $(1,-1,1)$

Sol:- Given that $\bar{F} = xy^2\bar{i} + 2x^2yz\bar{j} - 3yz^2\bar{k}$

Divergent $\bar{F} = \nabla \cdot \bar{F}$

2(b)

$$= \left(\bar{i}\frac{\partial}{\partial x} + \bar{j}\frac{\partial}{\partial y} + \bar{k}\frac{\partial}{\partial z}\right)\left(xy^2\bar{i} + 2x^2yz\bar{j} - 3yz^2\bar{k}\right)$$

$$= \frac{\partial}{\partial x}(xy^2) + \frac{\partial}{\partial y}(2x^2yz) + \frac{\partial}{\partial z}(-3yz^2)$$

$$= y^2 + 2x^2z - 6yz$$

$$\nabla \bar{F}_{(1,-1,1)} = 1 + 2 + 6 = 9$$

Ans: Given $\bar{F} = r^n \bar{r}$

We know that $x\bar{i} + y\bar{j} + z\bar{k}$

$$|\bar{r}| = \sqrt{x^2+y^2+z^2}$$

$$r^2 = x^2+y^2+z^2$$

$$\frac{\partial r}{\partial x} = x/r$$

$$\frac{\partial r}{\partial y} = y/r$$

$$\frac{\partial r}{\partial z} = z/r$$

$$\bar{F} = r^n \bar{r}$$

$$= r^n (x\bar{i} + y\bar{j} + \bar{k} z)$$

$$= r^n x\bar{i} + r^n y\bar{j} + r^n z\bar{k}$$

$$\text{div}(r^n \cdot \bar{r}) = \nabla \cdot (r^n \bar{r})$$

$$= \left(\bar{i} \cdot \frac{\partial}{\partial x} + \bar{j} \frac{\partial}{\partial y} + \bar{k} \frac{\partial}{\partial z}\right)\left(r^n x\bar{i} + r^n y\bar{j} + r^n z\bar{k}\right)$$

$$= \frac{\partial}{\partial x}(r^n x) + \frac{\partial}{\partial y}(r^n y) + \frac{\partial}{\partial z}(r^n z).$$

$$= \left[r^n \cdot 1 + x \cdot n \cdot r^{n-1} \frac{\partial r}{\partial x}\right] + \left[r^n \cdot 1 + y \cdot n \cdot r^{n-1} \frac{\partial r}{\partial y}\right]$$

$$+ \left[r^n \cdot 1 + z \cdot n \cdot r^{n-1} \frac{\partial r}{\partial z}\right]$$

$$= 3r^n + \left[x \cdot n \cdot r^{n-1} \frac{x}{r} + y n r^{n-1} \frac{y}{r} + z \cdot n r^{n-1} \frac{z}{r}\right]$$

$$= 3r^n + \left[x^2 n r^{n-2} + y^2 n r^{n-2} + z^2 n r^{n-2}\right]$$

$$= 3r^n + n r^{n-2} (x^2+y^2+z^2)$$

$$= 3r^n + n \cdot r^{n-2} r^2$$

$$= (3+n) r^n$$

If $r^n \cdot \bar{r}$ is solenoidal then

$$\text{div}(r^n \cdot \bar{r}) = 0$$

$$(3+n) r^n = 0$$

$$3 + n = 0$$

$$n = -3$$

$$\nabla \cdot \frac{\bar{r}}{r^3} = \left(\bar{i}\frac{\partial}{\partial x} + \bar{j}\frac{\partial}{\partial y} + \bar{k}\frac{\partial}{\partial z}\right)\frac{(x\bar{i} + y\bar{j} + z\bar{k})}{r^3}$$

$$= \left(\bar{i}\frac{\partial}{\partial x} + \bar{j}\frac{\partial}{\partial y} + \bar{k}\frac{\partial}{\partial z}\right)(x\bar{i} + y\bar{j} + \bar{k}z)\, r^{-3}$$

$$= \Sigma \frac{\partial}{\partial x}(x \cdot r^{-3})$$

$$= \Sigma \left[x \cdot -3 \cdot r^{-4}\frac{\partial r}{\partial x} + r^{-3}\right]$$

$$= \Sigma \left[-3x r^{-4}\frac{x}{r} + r^{-3}\right]$$

$$= \Sigma (-3r^{-5}x^2 + r^{-3})$$

$$= r^{-3} - 3r^{-5}x^2 + r^{-3} - 3y^2 r^{-5} + r^{-3} - 3z^2 r^{-5}$$

$$= 3r^{-3} - 3r^{-5}(x^2 + y^2 + z^2)$$

$$= 3r^{-3} - 3r^{-5}\, r^2$$

$$= 3r^{-3} - 3r^{-3} = 0$$

$$\nabla \cdot \frac{\bar{r}}{r^3} = 0$$

$\frac{\bar{r}}{r^3}$ is a solenoidal

Find the values of a & b so that the surface $ax^2 - byz = (a+2)x$ and $4x^2y + z^3 = 4$ may be intersect orthogonally at the point $(1,-1,2)$.

Sol:- The given surfaces are

$$ax^2 - byz = (a+2)x \quad \text{--- (1)}$$

$$4x^2y + z^3 = 4 \quad \text{--- (2)}$$

These two surfaces meet at the point $(1,-1,2)$

Sub a point in eqn (1) we get

$$a + 2b = a + 2$$

$$b = 1$$

let $\phi_1 = ax^2 - byz - (a+2)x$

$$\phi_2 = 4x^2y + z^3 - 4$$

$$\nabla\phi_1 = \bar{i}(2ax - a - 2) + \bar{j}(-bz) + \bar{k}(-by)$$

$$\nabla\phi_1(1,-1,2) = \bar{i}(2a - a - 2) - \bar{j}(2b) + \bar{k}(+b)$$

$$= \bar{i}(a-2) - 2\bar{j}b + b\bar{k}$$

$$\nabla\phi_1 = (a-2)\bar{i} - 2b\bar{j} + b\bar{k}$$

$$\nabla\phi_2 = \bar{i}(8xy) + \bar{j}(4x^2) + \bar{k}(3z^2)$$

$$\nabla\phi_2(1,-1,2) = -8\bar{i} + 4\bar{j} + 12\bar{k}$$

Since these two surfaces intersect orthogonally

then $\nabla\phi_1 \cdot \nabla\phi_2 = 0$

$$(\bar{i}(a-2) - 2b\bar{j} + b\bar{k})(-8\bar{i} + 4\bar{j} + 12\bar{k}) = 0$$

$$-8a + 16 - 8b + 12b = 0$$

$$-8a + 16 + 4b = 0$$

$$8a = 20$$

$$a = 20/8 = 5/2$$

$a = 5/2$, $b = 1$

21(b)

* If $\bar{f} = (x+y+1)\bar{i} + \bar{j} - (x+y)\bar{k}$ then show that $\bar{f} \cdot \text{Curl } \bar{f} = 0$.

Sol: Given that.

$$\bar{f} = (x+y+1)\bar{i} + \bar{j} - (x+y)\bar{k}$$

$$\text{Curl } \bar{f} = \nabla \times \bar{f}$$

$$= \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x+y+1 & 1 & -x-y \end{vmatrix}$$

$$= \bar{i}(-1-0) = \bar{j}(-1-0) + \bar{k}(0-1).$$

$$\text{Curl } \bar{F} = -\bar{i} + \bar{j} - \bar{k}.$$

$$\bar{f} = \text{Curl } \bar{f} = [(x+y+1)\bar{i} + \bar{j} - (x+y)\bar{k}][-\bar{i} + \bar{j} - \bar{k}]$$

$$= (x+y+1)(-1) + 1 + x + y = 0.$$

$$\boxed{= \bar{f} \text{ Curl } \bar{f} = 0}.$$

Sol:- Given that $\phi = 2x^3y^2z^4$

$$\nabla\cdot\phi = i\frac{\partial}{\partial x}(2x^3y^2z^4) + j\frac{\partial}{\partial y}(2x^3y^2z^4) + \bar{k}\frac{\partial}{\partial z}(2x^3y^2z^4)$$

5(a)

$$= i(6x^2y^2z^4) + \bar{j}(-8) + 32\bar{k}$$

$$= 24\bar{i} - 8\bar{j} + 32\bar{k}$$

And also given by $x^2yz = 3x + z^2$

The directional of normals to the surface

$$= \nabla\phi$$

let $f = xy^2z - 3x - z^2$

$$\nabla f = \bar{i}\frac{\partial f}{\partial x} + \bar{j}\frac{\partial f}{\partial y} + \bar{k}\frac{\partial f}{\partial z}$$

$$= \bar{i}\frac{\partial}{\partial x}(xy^2z - 3x - z^2) + \bar{j}\frac{\partial}{\partial y}(xy^2z - 3x - z^2) + \bar{k}\frac{\partial}{\partial z}(xy^2z - 3x - z^2)$$

$$= \bar{i}(y^2z - 3) + \bar{j}(2xyz) + \bar{k}(-2z + xy^2)$$

$$\nabla f(1,-2,1) = \bar{i} - 4\bar{j} + 2\bar{k} = \bar{a}\ (\text{say})$$

$$|\bar{a}| = \sqrt{1+16+4} = \sqrt{21}$$

Directional derivative $= \nabla\phi\ \dfrac{\bar{a}}{|\bar{a}|}$

$$= \frac{(24\bar{i} - 8\bar{j} + 32\bar{k})(\bar{i} - 4\bar{j} + 2\bar{k})}{\sqrt{21}}$$

$$= \frac{24 + 32 + 64}{\sqrt{21}} = \frac{120}{\sqrt{21}}$$

5(b) Prove that $\nabla\left[\nabla \cdot \bar{r}/r\right] = \frac{-2}{r^3}\bar{r}$

Sol:- We have $\nabla \cdot \frac{\bar{r}}{r} = \hat{i}\,\bar{i}\,\frac{\partial}{\partial x}\left(\frac{\bar{r}}{r}\right)$

$$= \hat{i}\,\bar{i}\left[\frac{1}{r}\frac{\partial \bar{r}}{\partial x} + \bar{r}\left(-\frac{1}{r^2}\right)\frac{x}{r}\right]$$

$$= \hat{i}\,\bar{i}\left[\frac{1}{r}\,\bar{i} - \frac{\bar{r}}{r^3}\,x\right]$$

$$= \frac{1}{r}\,\hat{i}\,\hat{i} - \frac{1}{r^3}\,r^2 = \frac{3}{r} - \frac{1}{r} = \frac{2}{r}$$

$$\nabla\left[\nabla \cdot \frac{\bar{r}}{r}\right] = \hat{i}\,\frac{\partial}{\partial x}\left(\frac{2}{r}\right)$$

$$= \hat{i}\left(\frac{-2}{r^2}\right)\left(\frac{\partial r}{\partial x}\right)$$

$$= \hat{i}\,\frac{-2}{r^3}\,x = \frac{-2}{r^3}\left(\hat{i}\,x\hat{i}\right) = \frac{-2\bar{r}}{r^3}$$

$$\nabla\left[\nabla \cdot \frac{\bar{r}}{r}\right] = \frac{-2\bar{r}}{r^3}$$

6) Find the constants a, b, c. Such that the vector $\bar{A} = (x+2y+az)\bar{i} + (bx-3y-z)\bar{j} + (4x+cy+2z)\bar{k}$ is irrotational. such that $\bar{A} = \nabla\phi$.

Sol:- Given $\bar{A} = (x+2y+az)\bar{i} + (bx-3y-z)\bar{j} + (4x+cy+2z)\bar{k}$ — ①

and $\bar{A} = \nabla \times \bar{A}$

$$= \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x+2y+az & bx-3y-z & 4x+cy+2z \end{vmatrix}$$

$$= \bar{i}(c+1) - \bar{j}(4-a) + \bar{k}(b-2)$$

$$\nabla \times \bar{A} = (c+1)\bar{i} + (a-4)\bar{j} + (b-2)\bar{k}$$

if $\bar{A}$ is irrotational.

$$\nabla \times \bar{A} = \bar{0}$$

$$(c+1)\bar{i} + (a-4)\bar{j} + (b-2)\bar{k} = 0\bar{i} + 0\bar{j} + 0\bar{k}$$

$c+1 = 0 \Rightarrow c = -1$

$a-4 = 0 \Rightarrow a = 4$

$b-2 = 0 \Rightarrow b = 2$

a, b, c values substitute in eqn ①

$$\bar{A} = (x+2y+4z)\bar{i} + (2x-3y-z)\bar{j} + (4x-y+2z)\bar{k}$$

We know. $\bar{A} = \nabla\phi$

$$(x+2y+4z)\bar{i} + (2x-3y-z)\bar{j} + (4x-y+2z)\bar{k}$$
$$= \bar{i}\frac{\partial\phi}{\partial x} + \bar{j}\frac{\partial\phi}{\partial y} + \bar{k}\frac{\partial\phi}{\partial z}$$

$$\frac{\partial\phi}{\partial x} = x+2y+4z$$

$$\frac{\partial\phi}{\partial y} = 2x-3y-z$$

$$\frac{\partial\phi}{\partial z} = 4x-y+2z$$

$$\int d\phi = \int (x+2y+4z)\,dx + (2x-3y-z)\,dy + (4x-y+2z)\,dz$$

$$\phi = \frac{x^2}{2} + \int 2y\,dx + \int 4z\,dx + \int 2x\,dy - \frac{3y^2}{2} - \int z\,dy + \int 4x\,dz$$

$$- \int y\,dz + \frac{2z^2}{2} + C$$

$$= \frac{x^2}{2} - \frac{3y^2}{2} + z^2 + 2xy + 4xz + 2xy - yz + 4xz$$

$$- zy + C$$

$$= \frac{x^2}{2} - \frac{3y^2}{2} + z^2 + 4xy - 2yz + 8xz + C$$

* Show that curl $(r^n \vec{r}) = 0$

Sol:- Given that $r^n \vec{r} = 0$

7(a)

$$\vec{r} = x\bar{i} + y\bar{j} + z\bar{k}$$

$$r = |\vec{r}| = \sqrt{x^2+y^2+z^2}$$

$$r^2 = x^2+y^2+z^2$$

$$\frac{\partial r}{\partial x} = \frac{x}{r}, \quad \frac{\partial r}{\partial y} = \frac{y}{r}, \quad \frac{\partial r}{\partial z} = \frac{z}{r}.$$

$$\therefore r^n \vec{r} = r^n x\bar{i} + r^n y\bar{j} + r^n z\bar{k}$$

$$\text{curl}(r^n \vec{r}) = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ r^n x & r^n y & r^n z \end{vmatrix}$$

$$= \Sigma \bar{i}\left[\frac{\partial}{\partial y}(r^n z) - \frac{\partial}{\partial z}(r^n y)\right]$$

$$= \Sigma \bar{i}\left(z \cdot n \cdot r^{n-1}\frac{\partial r}{\partial y} - y n r^{n-1}\frac{\partial r}{\partial z}\right)$$

$$= n r^{n-1} \Sigma \bar{i}\left(z\frac{y}{r} - y\frac{z}{r}\right)$$

$$= n \cdot r^{n-1} \Sigma \bar{i}(0)$$

$$= 0\bar{i} + 0\bar{j} + 0\bar{k}$$

$$\Sigma r^n \vec{r} = \bar{0}$$

7(b) $\nabla^2[f(r)] = f''(r) + \frac{2}{r} f'(r)$. (OR) Show that

$\text{div}[\text{grad } f(r)] = f''(r) + \frac{2}{r} f'(r)$

We know that,

the position vector $\bar{r} = x\bar{i} + y\bar{j} + z\bar{k}$

let $r = |\bar{r}| = \sqrt{x^2+y^2+z^2}$

$r^2 = x^2+y^2+z^2$

$\frac{\partial r}{\partial x} = \frac{x}{r}$

$\frac{\partial r}{\partial y} = \frac{y}{r}$

$\frac{\partial r}{\partial z} = \frac{z}{r}$

$\text{grad}[f(r)] = \nabla f(r)$

$= \sum \bar{i} \frac{\partial}{\partial x}(f(x))$

$= \sum \bar{i} f'(r) \frac{\partial r}{\partial x}$

$= \sum \bar{i} f'(r) \frac{x}{r}$

$\text{grad}[f(r)] = \sum i \frac{x f'(r)}{r}$

$\text{div}[\text{grad } f(r)] = \nabla \sum i \frac{x f'(r)}{r}$

$= \sum \frac{\partial}{\partial x}\left[\frac{x f'(r)}{r}\right]$

$= \sum \frac{r[x f''(r) \partial r/\partial x + f'(r)1] - x f'(r) \frac{\partial r}{\partial x}}{r^2}$

$= \sum \frac{r\left[\frac{x^2 f''(r)}{r} + f'(r)\right] - \frac{x^2}{r} f(r)}{r^2}$

$= \sum \frac{x^2 f''(r) + r f'(r) - \frac{x^2}{r} f'(r)}{r^2}$

$= \frac{f''(r) \cdot \sum x^2 + 3r f'(r) + \frac{1}{r} f'(r) \cdot r^2}{r^2}$

$$= f''(x) + \frac{3}{x} f'(x) - \frac{1}{x} f'(x)$$

$$= f''(x) + \frac{2}{x} f'(x)$$

$$\nabla^2 [f(x)] = f''(x) + \frac{x}{2} f'(x)$$

$$\text{div}\,[\text{grad}\ f(x)] = f''(x) + \frac{x}{2} f(x)$$

8(a)

* Prove that $\text{grad}(fg) = f(\text{grad } g) + g(\text{grad } f)$.

Sol:- LHS

$$\text{grad}(fg) = \nabla(fg)$$

$$= \left(\bar{i}\frac{\partial}{\partial x} + \bar{j}\frac{\partial}{\partial y} + \bar{k}\frac{\partial}{\partial z}\right)(fg)$$

$$= \sum \bar{i}\frac{\partial}{\partial x}(fg)$$

$$= \sum \bar{i}\left(g\frac{\partial f}{\partial x} + f\frac{\partial g}{\partial x}\right)$$

$$= \bar{i}\left(g\frac{\partial f}{\partial x} + f\frac{\partial g}{\partial x}\right) + \bar{j}\left(g\frac{\partial f}{\partial y} + f\frac{\partial g}{\partial y}\right) + \bar{k}\left(g\frac{\partial f}{\partial z} + f\frac{\partial g}{\partial z}\right)$$

$$= f\left(\bar{i}\frac{\partial g}{\partial x} + \bar{j}\frac{\partial g}{\partial x} + \bar{k}\frac{\partial g}{\partial k}\right) + g\left(\bar{i}\frac{\partial f}{\partial x} + \bar{j}\frac{\partial f}{\partial y} + \bar{k}\frac{\partial f}{\partial z}\right)$$

$$= f(\nabla g) + g(\nabla f) = f$$

$$= f(\text{grad } g) + g(\text{grad } f) = \text{RHS}$$

$\therefore$ LHS = RHS $\Rightarrow$ $\text{grad}(fg) = f(\text{grad } g) + g(\text{grad } f)$

2(b)

Find curl $\bar{F}$ where $\bar{F} = \text{grad}(x^3+y^3+z^3-3xyz)$

Sol:- Given that,

$$\bar{F} = \text{grad}(x^3+y^3+z^3-3xyz).$$

$$\nabla(x^3+y^3+z^3-3xyz).$$

$$= \left(\bar{i}\frac{\partial}{\partial x} + \bar{j}\frac{\partial}{\partial y} + \bar{k}\frac{\partial}{\partial z}\right)(x^3+y^3+z^3-3xyz).$$

$= \bar{i}(3x^2 - 3yz) + \bar{j}(3y^2 - 3xz) + \bar{k}(3z^2 - 3xy)$

Curl $\bar{F} = \nabla \times \bar{F}$.

$$= \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 3x^2 - 3yz & 3y^2 - 3xz & 3z^2 - 3xy \end{vmatrix}$$

$= \bar{i}(-3x + 3x) - \bar{j}(-2y + 3y) + \bar{k}(-3z + 3z)$.

$= 0\bar{i} + 0\bar{j} + 0\bar{k}$.

$= \bar{0}$.

$\therefore \boxed{\text{Curl } \bar{F} = \nabla \times \bar{F} = \bar{0}}$.

9 ~~*~~ If $f = (x^2+y^2+z^2)^{-n}$ then find div grad f and determined n if div grad $f = 0$

Sol:- Given $f = (x^2+y^2+z^2)^{-n}$

let $\bar{r} = x\bar{i} + y\bar{j} + z\bar{k}$

$$r = |\bar{r}| = \sqrt{x^2+y^2+z^2}$$

$$r^2 = x^2+y^2+z^2$$

$$f(r) = (r^2)^{-n}$$

$$f'(r) = -2n\, r^{-2n-1}$$

$$f''(r) = (-2n)(-2n-1)\, r^{-2n-2} = 2n(2n+1)\, r^{-2n-2}$$

We have div grad $f = \nabla^2 f(r) = f''(r) + \frac{2}{r} f'(r)$

$$= (2n)(2n+1)\, r^{-2n-1} - 4n\, r^{-2n-1}$$

$$= r^{-2n-2}\left[2n(2n+1-2)\right]$$

$$\text{div grad } f = \nabla^2 f(r) = 2n(2n-1)\, r^{-2n-2}$$

If div grad $f = 0$

$$(2n)(2n-1)\, r^{-2n-2} = 0$$

(b) $n = 0$ or $n = 1/2$.

10(a) Show that the vector $(x^2-yz)\bar{i}+(y^2-zx)\bar{j}+(z^2-xy)\bar{k}$ is irrotational and find its scalar potential.

Let $\bar{f}=(x^2-yz)\bar{i}+(y^2-zx)\bar{j}+(z^2-xy)\bar{k}$

Curl $\bar{F}_1 = \nabla \times \bar{F}$

$$= \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2-yz & y^2-xz & z^2-xy \end{vmatrix}$$

$$= \bar{i}(-x+x) - \bar{j}(-y+y) + \bar{k}(-z+z)$$

$$= 0$$

$$\nabla \times \bar{F} = 0$$

$\bar{F}$ is irrotational.

$$\bar{F} = \nabla \phi$$

$$(x^2-yz)\bar{i} + (y^2-zx)\bar{j} + (z^2-xy)\bar{k} = \bar{i}\frac{\partial \phi}{\partial x} + \bar{j}\frac{\partial \phi}{\partial y} + \bar{k}\frac{\partial \phi}{\partial z}$$

$$\frac{\partial \phi}{\partial x} = x^2 - yz$$

P.O.B.S w.r.t 'x'

$$\phi = \frac{x^3}{3} - xyz + c_1 \quad \text{--- (1)}$$

$$\frac{\partial \phi}{\partial y} = y^2 - xz$$

P.O.B.S w.r.t 'x'

$$\phi = \frac{y^2}{3} - xyz + c_2 \quad \text{--- (2)}$$

$$\frac{\partial \phi}{\partial z} = z^2 - xy$$

$$\phi = \frac{z^3}{3} - xyz + c_3 \quad \text{--- (3)}$$

Adding (1), (2) & (3)

$$\left(\cancel{\phi + \phi + \phi = \frac{x^3}{3} - xyz + c_1}\right) \times$$

$$\phi = \frac{x^3}{3} + \frac{y^3}{3} + \frac{z^3}{3} - xyz + c$$

Which is the required scalar Potential.

10(b) Curl $F = \bar{i} + \bar{j} + \bar{k}$.

* Find curl $\bar{F}$ at the $(1,-1,1)$ where $F = xy^2\bar{i} + 2x^2yz\,\bar{j} - 3yz^2\bar{k}$.

Sol:- Given that

$F = xy^2\bar{i} - 2x^2yz\bar{j} - 3yz^2\bar{k}$

$$\text{Curl } F = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xy^2 & -2x^2yz & -3yz^2 \end{vmatrix}$$

$$= \bar{i}(-3z^2 - 2x^2y) + \bar{j}(0-0) + \bar{k}(4xyz - 2xy$$

$$= (-3z^2 - 2x^2y)\bar{i} + (4xyz - 2xy)\bar{k}$$

$$\nabla \times \bar{F}(1,-1,1) = (-3+2)\bar{i} + (-4+2)\bar{k}$$

$$= -\bar{i} - 2\bar{k}$$

(11) Find a directional derivative of $f = x^2 - y^2 + 2z^2$ at the point $(1,2,3)$ in the direction of the line (P, Q) where Q is the point $(5,0,4)$ Now also calculate the magnitude of the maximum directional derivative.

Given $f = x^2 - y^2 + 2z^2$

$$\nabla f = \bar{i}\frac{\partial f}{\partial x} + \bar{j}\frac{\partial f}{\partial y} + \bar{k}\frac{\partial f}{\partial z}$$

$$= \bar{i}\frac{\partial}{\partial x}(x^2 - y^2 + 2z^2) + \bar{j}\frac{\partial}{\partial y}(x^2 - y^2 + 2z^2) + \bar{k}\frac{\partial}{\partial z}(x^2 - y^2 + 2z^2)$$

$$= \bar{i}(2x) + \bar{j}(-2y) + \bar{k}(4z)$$

$$\nabla f_{(1,2,3)} = 2\bar{i} - 4\bar{j} + 12\bar{k}$$

and also given in the direction of line (P, Q)

$$P_{(1,2,3)} \Rightarrow \overline{OP} = \bar{i} + 2\bar{j} + 3\bar{k}$$

$$Q_{(5,0,4)} \Rightarrow \overline{OQ} = 5\bar{i} + 4\bar{k}$$

$$\overline{PQ} = \overline{OQ} - \overline{OP}$$

$$= 4\bar{i} - 2\bar{j} + \bar{k} = \bar{a}$$

$$|\bar{a}| = \sqrt{16+4+1} = \sqrt{21}$$

Directional derivative $= \nabla f \cdot \frac{\bar{a}}{|\bar{a}|}$

$$= (2\bar{i} - 4\bar{j} + 12\bar{k})\frac{(4\bar{i} - 2\bar{j} + \bar{k})}{\sqrt{21}} = \frac{28}{\sqrt{21}}$$

D.D is maximum in the direction of ∇f

Hence, Maximum D.D in the direction of

$2\bar{i} - 4\bar{j} + 12\bar{k}$.

Magnitude $= \sqrt{4+16+144} = \sqrt{164} = 2\sqrt{41}$

12/ Show that $\operatorname{div}\left[\operatorname{grad} r^m\right] = m(m+1)\, r^{m-2}$, $\bar{r} = x\bar{i} + y\bar{j} + z\bar{k}$, $r = |\bar{r}|$.

(or) $\nabla(\nabla r^m) = m(m+1)\, r^{m-2}$

$\nabla(\nabla r^n) = n(n+1)\, r^{n-2}$

Sol:- Given that $\bar{r} = x\bar{i} + y\bar{j} + z\bar{k}$

$$|\bar{r}| = \sqrt{x^2+y^2+z^2} = r^2 = |\bar{r}| = x^2+y^2+z^2$$

$$\frac{\partial r}{\partial x} = \frac{x}{r}, \quad \frac{\partial r}{\partial y} = \frac{y}{r}, \quad \frac{\partial r}{\partial z} = \frac{z}{r}$$

$$\operatorname{grad} r^m = \nabla r^m = \left(\bar{i}\frac{\partial}{\partial x} + \bar{j}\frac{\partial}{\partial y} + \bar{k}\frac{\partial}{\partial z}\right) r^m$$

$$= \sum i \frac{\partial (r^m)}{\partial x} = \sum 1 \cdot m \cdot r^{m-1} \frac{\partial r}{\partial x}$$

$$= \xi\, i\, m \cdot r^{m-1} \frac{x}{r}$$

$$\text{grad } r^m = \nabla r^m = \xi\, i\, m\, r^{m-2} x$$

$$\text{div}\left[\text{grad } r^m\right] = \nabla \cdot (\nabla r^m)$$

$$= \xi \frac{\partial}{\partial x} \left(m r^{m-2} x\right)$$

$$= m\, \xi \frac{\partial}{\partial x} \left(r^{m-2} x\right)$$

$$= m\, \xi \left[r^{m-2} 1 + x (m-2) r^{m-3} \frac{\partial r}{\partial x}\right]$$

$$= m\, \xi \left[r^{m-2} + (m-2)\, r^{m-4} x^2\right]$$

$$= m \left[\xi \left[\left[\xi\, r^{m-1}\right] + (m-2) r^{m-4}\, \xi\, x^2\right]\right]$$

$$= m \left[3 r^{m-2} + (m-2)\, r^{m-2}\right]$$

$$= m\, r^{m-2} \left[3 + m - 2\right] = m (m+1)\, r^{m-2}$$